

Emily Herbert

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Design Engineer focused on developer-facing interfaces, interaction systems, and tools for understanding complex software.

Education

- 2021 **MS in Computer Science**, *The University of Massachusetts Amherst*, Amherst, MA.
- 2018 **BS in Computer Science**, *Trinity University*, San Antonio, TX.
- 2025 **Art & Design Intensive**, *Atelier de Sevres*, Paris, France.

Relevant Experience

- Oct 2025 – **TAHO (early-stage startup)**.
- Apr 2026 Product Engineer - Rust
 - Deployed large-scale models (Qwen3) on commodity hardware under memory/latency constraints, validated up to 32B parameters.
 - Built distributed inference components using streaming-first decomposition to enable execution without full model materialization.
 - Investigated limitations in inference frameworks (especially ONNX in Rust) and synthesized constraints to guide system design.
 - Built tooling (graph visualization, TUI) used in team workflows to debug and inspect model execution.
 - Integrated PyTorch, ONNX, and onnxruntime to bridge model formats and execution runtimes.
- Aug 2021 - **Fuel Labs (early-stage startup)**.
- May 2023 Compiler Engineer - Rust
 - Contributed to the Sway smart contract language and compiler, used by a large developer ecosystem (61k+ [GitHub](#) stars).
 - Implemented core language features (match expressions, traits, generics) informed by programming language research.
 - Translated developer feedback into language and tooling improvements.
 - Delivered technical talks and demos to onboard developers and explain system design.
- May 2021 – **Google, Madison, WI**.
- Aug 2021 Software Engineering Intern - C++
 - Implemented load balancing in a high-performance networking library using RMA-based RPC mechanisms.
 - Integrated low-level systems for efficient communication with network hardware.
- June 2018 - **Northeastern University, University of Massachusetts Amherst, Boston, MA, Amherst, MA**.
- May 2021 PhD Researcher - Rust, TypeScript, Python
 - Built a serverless function accelerator compiling traced JavaScript into Rust to improve latency and throughput.
 - Led development across compiler, runtime, networking, and storage layers in a $0 \rightarrow 1$ system.
 - Implemented logging and debugging systems to diagnose non-deterministic behavior under limited observability.
 - Built and trained neural network models using TensorFlow/Keras with data preprocessing and evaluation pipelines.
- June 2017 – **National Aeronautics and Space Administration (NASA), Langley, VA**.
- Aug 2017 Software Engineering Intern - C++
 - Contributed to GPS-based geofencing system for autonomous drones across heterogeneous platforms.
 - Tested and validated system behavior using simulated hardware environments.

Selected Talks & Publications

- 2023 Blockchain Language Design panel. *Layer 2 Day at EthDenver*. [[recording](#)]
- 2022 Developing Smart Contracts in Sway talk. *Layer2Amsterdam*. [[slides](#)]
Sway: A Rust-based Smart Contract Language talk. *EthDenver*. [[slides](#)]
- 2021 Emily Herbert and Arjun Guha. A Language-based Serverless Function Accelerator. [[preprint](#), [repo](#)]
- 2020 Wang Cen, Emily A Herbert, and Peter J Haas. NIM: Modeling and Generation of Simulation Inputs via Generative Neural Networks. *Winter Simulation Conference*. [[paper](#)]. **Best Contributed Theoretical Paper Finalist**.

Skills

Core Languages, Rust (primary), TypeScript, Python, Haskell, Scala, Sway .

AI/ML, PyTorch, ONNX, onnxruntime, TensorFlow/Keras, NumPy, Hugging Face .

Systems, Distributed systems, runtime design, IR design, low-latency systems, Protobuf, async (Tokio), rkyv .

Tools, Docker, Kubernetes, AWS (EC2, Lambda, S3), GCP, React, TUI (ratatui), CLI (clap), VSCode, Claude, ChatGPT .